

3 METHOD

3.1 PROBLEM FORMULATION

Let $\mathcal{R}$ and $\mathcal{E}$ denote reaction and enzyme sets. Observed associations form a many-to-many bipartite graph $\mathcal{D} \subseteq \mathcal{R} \times \mathcal{E}$. We learn enzyme–reaction compatibility by independently mapping both modalities to a shared space. The complete reaction and enzyme encoders f_R, f_E give normalized outputs $\mathbf{z}_r = f_R(r)$ and $\mathbf{z}_e = f_E(e)$, yielding the cosine score $S(r, e) = \mathbf{z}_r^\top \mathbf{z}_e \in [-1, 1]$.

The same score supports $E \rightarrow R$ retrieval over reaction candidates $\mathcal{C}_R(e) \subseteq \mathcal{R}$ and $R \rightarrow E$ retrieval over enzyme candidates $\mathcal{C}_E(r) \subseteq \mathcal{E}$. Relevant candidates are each query’s observed neighbors in the test graph, allowing multiple correct results. Query sets, candidate pools, and positive neighborhoods can differ between directions. Learning uses only the training graph $\mathcal{D}_{\text{tr}} \subseteq \mathcal{R}_{\text{tr}} \times \mathcal{E}_{\text{tr}}$.

V4 first aligns the encoders, then refines their embeddings over the complete training graph before adding training–reaction coordinates.

3.2 ENZYME REPRESENTATION

Catalytic function depends on both sequence context and functional residues. We capture these signals using frozen ProtT5 residue embeddings $\mathbf{H}_e \in \mathbb{R}^{L \times 1024}$ (Elnaggar et al., 2022) and scores from a frozen SLEEC model (Gong et al., 2026), complemented by K learned residue summaries. The global ProtT5 feature is $\mathbf{x}_e^P = L^{-1} \sum_{i \in \mathcal{V}_e} \mathbf{h}_i$, where $\mathcal{V}_e$ indexes the L valid residues.

SLEEC-guided pooling. We suppress e on residue vectors and attention weights. SLEEC assigns each $\mathbf{h}_i \in \mathbb{R}^{1024}$ a logit ℓ_i , centered and clipped as $\tilde{\ell}_i = \text{clip}(\ell_i - L^{-1} \sum_{j \in \mathcal{V}_e} \ell_j, -10, 10)$. The SLEEC feature $\mathbf{x}_e^S \in \mathbb{R}^{1024}$ summarizes enzyme e by averaging its residue embeddings with normalized SLEEC weights:

$$a_i^S = (1 - \eta) \frac{\exp(\tilde{\ell}_i)}{\sum_{j \in \mathcal{V}_e} \exp(\tilde{\ell}_j)} + \frac{\eta}{L}, \quad \mathbf{x}_e^S = \sum_{i \in \mathcal{V}_e} a_i^S \mathbf{h}_i, \quad (1)$$

Here $\eta = 0.05$ retains a uniform contribution from every valid residue.

Learned residue views. We learn queries $\mathbf{q}_k \in \mathbb{R}^{256}$, $k = 1, \dots, K$, shared across enzymes. A shared residue adapter and separate key and value projections produce normalized keys $\mathbf{k}_i$ and unnormalized values $\mathbf{v}_i$ in $\mathbb{R}^{256}$. Write $\mathbf{n}(\mathbf{x}) = \mathbf{x} / \max(\|\mathbf{x}\|_2, \epsilon)$ for numerical normalization, with $\epsilon = 10^{-6}$ in this encoder. Query $\hat{\mathbf{q}}_k = \mathbf{n}(\mathbf{q}_k)$ defines

$$p_{ki} = \frac{\exp(\gamma \hat{\mathbf{q}}_k^\top \mathbf{k}_i)}{\sum_{j \in \mathcal{V}_e} \exp(\gamma \hat{\mathbf{q}}_k^\top \mathbf{k}_j)}, \quad \mathbf{x}_e^k = \sum_{i \in \mathcal{V}_e} \left[(1 - \eta) p_{ki} + \frac{\eta}{L} \right] \mathbf{v}_i, \quad (2)$$

Thus p_{ki} is a scalar weight over residues for query k ; the learned scale $\gamma = 10 \text{ sigmoid}(\rho_\gamma) \in (0, 10)$ controls concentration. Each view combines 95% attention-weighted and 5% uniform pooling of the values. Phase 1 trains the queries, adapter, and projections; frozen views can be cached per enzyme for reuse across candidate reactions. We retain $K = 4$ as a compromise across screening and retrieval: the single-seed $K \in \{1, 2, 4, 8\}$ ablation finds no uniformly best count (Appendix J).

Feature-wise gated fusion. We combine the extracted enzyme features $\{\mathbf{x}_e^P, \mathbf{x}_e^S\} \cup \{\mathbf{x}_e^k\}_{k=1}^K$ by projecting each feature into a common 256-dimensional space using an independent MLP followed by ℓ_2 normalization. The projected features are $\mathbf{y}_e^v = \mathbf{n}(F_v(\mathbf{x}_e^v))$, with view labels $v \in \mathcal{I} = \{P, S, 1, \dots, K\}$. A gating network maps their concatenation to feature-specific weights $\omega_{e, vd}$, normalized across views for each coordinate d . The fused representation is

$$\bar{\mathbf{y}}_{e, d} = \sum_{v \in \mathcal{I}} \omega_{e, vd} \mathbf{y}_{e, d}^v, \quad d = 1, \dots, 256. \quad (3)$$

Independent projections of $\mathbf{x}_e^P$ and $\bar{\mathbf{y}}_e$ yield normalized 512-dimensional vectors $\mathbf{g}_e$ and $\mathbf{f}_e$. The enzyme embedding is $\mathbf{b}_e = \mathbf{n}(\mathbf{g}_e + s \mathbf{f}_e)$, where the learned scalar $s \in (0, 0.5)$ controls the fused contribution (Appendix A.2).

3.3 REACTION ENCODER

The reaction encoder combines frozen ReactionT5v2, Uni-Mol2, and ChIRo features (Sagawa & Kojima, 2023; Ji et al., 2024; Adams et al., 2022) with a 617-dimensional chemistry descriptor computed from participant SMILES. This descriptor concatenates averaged Morgan fingerprints (512 coordinates), four aggregates of 16 molecular properties (64), participant-set statistics (4), and cofactor indicators (37). It supplies explicit substructure, physicochemical, and cofactor cues alongside pretrained embeddings. "Fixed" means its construction is not optimized during retrieval training; property normalization and the cofactor vocabulary are fitted on training reactions and frozen; its neural projection is learned (Appendix D).

For each molecular modality $m \in \{\text{UniMol2}, \text{ChIRo}\}$, independent attention poolers summarize reactant and product embeddings as weighted means $\mathbf{p}_-^m$ and $\mathbf{p}_+^m$. Their concatenation $\mathbf{c}_r^m = [\mathbf{p}_-^m; \mathbf{p}_+^m; \mathbf{p}_+^m - \mathbf{p}_-^m; |\mathbf{p}_+^m - \mathbf{p}_-^m|]$ retains both summaries alongside their signed and absolute differences. Independent MLPs map these two compositions, the ReactionT5v2 embedding, and the chemistry descriptor to four 512-dimensional tokens. Modality attention averages available tokens with learned weights; an output MLP and ℓ_2 normalization yield $\mathbf{b}_r$.

EnzymeMap supplies physical reactant and product sides. ReactZyme supplies an unordered participant set, copied to both poolers. Their parameters differ, so $\mathbf{p}_-^m$ and $\mathbf{p}_+^m$ can differ without representing an observed reaction direction. The chemistry descriptor combines participants irrespective of side. Appendix A.3 specifies the encoder.

3.4 ANCHOR-BALANCED MULTI-POSITIVE ALIGNMENT

We first align the encoders with a decoupled multi-positive objective, accounting for the many-to-many structure of enzyme–reaction associations. For anchor a , $P_B(a)$ contains all known training neighbors among the batch’s distinct opposite-modality endpoints; $N_B(a)$ contains the remainder.

With $s_{ax}^{(1)} = \mathbf{b}_a^\top \mathbf{b}_x / \tau_1$ and $\tau_1 = 0.2$, we minimize

$$\mathcal{L}_{\text{align}}^{(1)} = \frac{1}{2} \sum_{X \in \{R, E\}} \frac{1}{|\mathcal{A}_X|} \sum_{a \in \mathcal{A}_X} \frac{1}{|P_B(a)|} \sum_{p \in P_B(a)} \log \left(1 + \sum_{n \in N_B(a)} e^{s_{an}^{(1)} - s_{ap}^{(1)}} \right), \quad (4)$$

where $\mathcal{A}_X$ contains type- X anchors with nonempty positive and negative sets. Each positive competes only with $N_B(a)$, excluding other known positives from its denominator. Normalizing by $|P_B(a)|$ prevents anchors with more positives from receiving greater weight; the two directional losses are averaged equally. The implemented phase-1 objective also includes entropy and diversity penalties on residue attention (Appendix A.4).

3.5 FULL-GRAPH REFINEMENT

After initial alignment, we freeze both encoders and refine their representations over the complete training association graph. Two residual heads, h_E and h_R , learn additive adjustments to enzyme and reaction embeddings. For endpoint a of modality $X \in \{R, E\}$, the refined representation is $\mathbf{u}_a^{\text{tr}} = \mathbf{n}(\mathbf{b}_a + \kappa h_X(\mathbf{b}_a))$, where $\kappa = 0.2$ controls the residual contribution. We optimize the heads using cross-entropy with a uniform target distribution over each query’s known positive partners:

$$\mathcal{L}_{\text{CE}}^{(2)} = \frac{1}{2} \sum_{X \in \{R, E\}} \frac{1}{|\mathcal{T}_X|} \sum_{a \in \mathcal{T}_X} \left[\log \sum_{x \in \mathcal{T}_{\bar{X}}} e^{s_{ax}^{(2)}} - \frac{1}{|P(a)|} \sum_{p \in P(a)} s_{ap}^{(2)} \right], \quad (5)$$

where $\mathcal{T}_R = \mathcal{R}_{\text{tr}}$, $\mathcal{T}_E = \mathcal{E}_{\text{tr}}$, $\bar{X}$ denotes the opposite modality, and $P(a)$ contains the training positives of a . Scores are $s_{ax}^{(2)} = (\mathbf{u}_a^{\text{tr}})^\top \mathbf{u}_x^{\text{tr}} / \tau_2$, with $\tau_2 = 0.2$. Each query is compared with every opposite-modality training endpoint, including its positive partners. An identity penalty $\mathcal{L}_{\text{id}}$ discourages cosine displacement from the base embeddings, averaged equally over reactions and enzymes. The complete refinement objective is

$$\mathcal{L}^{(2)} = \mathcal{L}_{\text{CE}}^{(2)} + 10\mathcal{L}_{\text{id}}. \quad (6)$$

Only the residual heads are updated; both endpoint representations remain independently computable. Appendix A.5 gives the head architecture and identity-penalty definition.

3.6 BIDIRECTIONAL SCORING

A dictionary indexes training reactions. Frozen features are normalized, centered on training endpoints, and renormalized. Reaction coordinates $\mathbf{a}_r$ are normalized exponential cosine similarities to training reactions. For each enzyme, we find the 32 nearest training proteins; each reaction coordinate takes the maximum exponential similarity among linked neighbors. Normalization gives $\mathbf{a}_e$. Both kernels use temperature 0.03 (Appendix A.7).

For retrieval, we increase the fused protein contribution to $\tilde{\mathbf{b}}_e = \mathbf{n}(\mathbf{g}_e + 3s\mathbf{f}_e)$ and halve the residual coefficient, obtaining $\mathbf{u}_e = \mathbf{n}(\tilde{\mathbf{b}}_e + 0.1h_E(\tilde{\mathbf{b}}_e))$ and $\mathbf{u}_r = \mathbf{n}(\mathbf{b}_r + 0.1h_R(\mathbf{b}_r))$. These adjustments follow training on the native $\mathbf{b}_e, \mathbf{b}_r$. For either endpoint x , the complete representation and score are

$$\mathbf{z}_x = [\sqrt{0.6}\mathbf{u}_x; \sqrt{0.4}\mathbf{a}_x], \quad S(r, e) = \mathbf{z}_r^\top \mathbf{z}_e = 0.6\mathbf{u}_r^\top \mathbf{u}_e + 0.4\mathbf{a}_r^\top \mathbf{a}_e. \quad (7)$$

The $512 + |\mathcal{R}_{\text{tr}}|$ dimensional vectors support both directions without direction-specific heads or reranking. Each target training split has independently fitted retrieval weights and dictionary statistics (Appendix H).